

Adventure Works – Dashboard

Dashboard Documentation

Project: AW Sales

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1. Objectives

The main objectives of the Adventure Works dashboard are:

- Provide a comprehensive overview of sales performance, key metrics, and revenue trends to support data-driven decision-making.
- Enable detailed analysis of sales data by product, salesperson, and geographical location to identify top performers and market opportunities.
- Facilitate strategic planning by analyzing customer demographics and product sales to better understand market demand and optimize marketing efforts.

2. Semantic Model & BI Transformations

2.1. Data Sources

From the data warehouse, 8 tables are imported into Power BI:

- dim_costumer
- dim_date
- dim_product
- dim_salesperson
- dim_ship_method
- dim_territory
- fact_sales_order
- fact_sales_summary_monthly

The dashboard contains two panels, “Sales Overview” and “Sales Team Details”. Both panels use all the available tables. More about these tables and their relationship is presented in the [section below](#).

Data Update Schedule

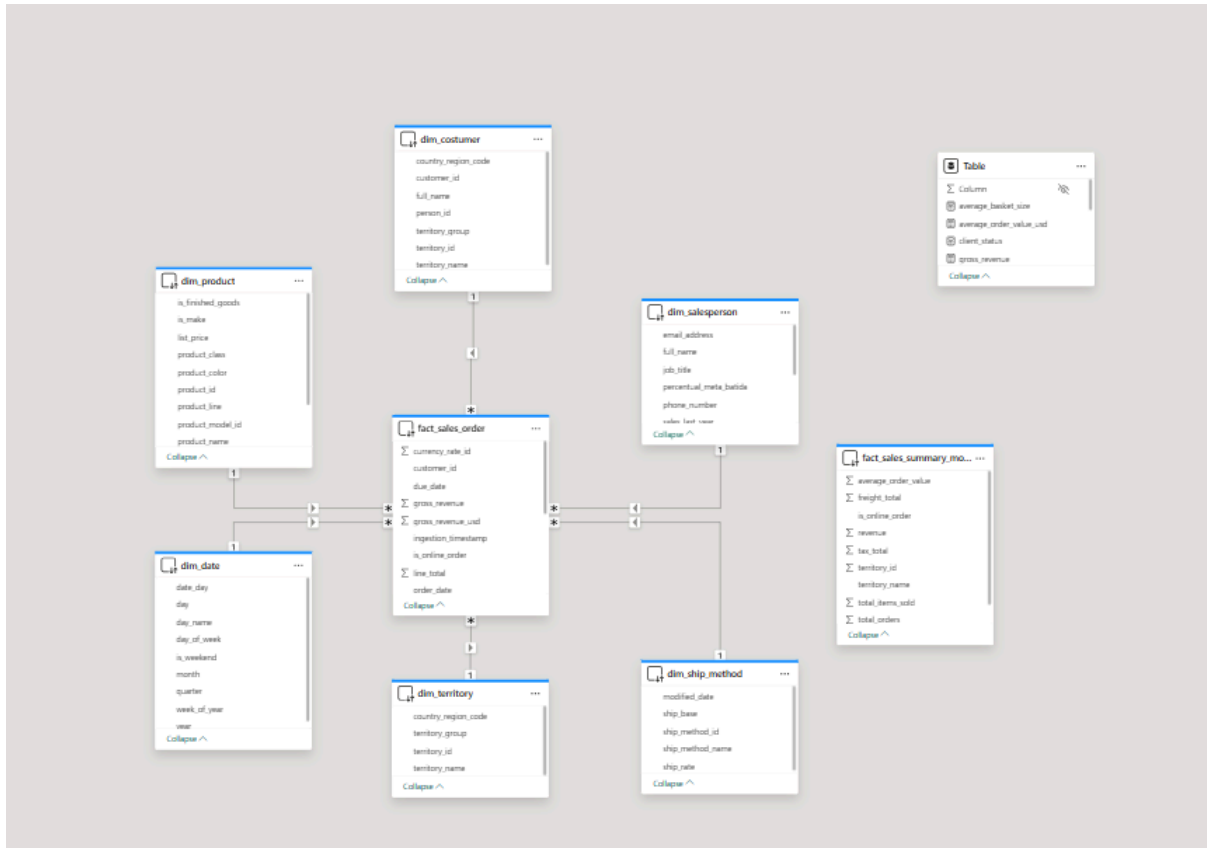
The dashboard is set to be updated daily at 21:10 PM (UTC).

Power Query Processing

Once the tables are loaded into Power BI through Power Query, no transformations are made inside of Power Query. All the transformations were concentrated on dbt.

2.2. Semantic Model

The data inside Power BI is structured as shown in the picture below.



In total, there are eight tables on the Adventure Works semantic model, two facts tables, 6 dims and one OBT. There are also four support virtual tables with measures created inside Power BI. These metrics will be explained later in this document.

The following table presents each data object with a brief description and their corresponding keys:

Table	Description
fact_sales_order	This fact table records individual sales transactions with key metrics.
fact_sales_summary_monthly	This aggregated fact table summarizes sales data on a monthly basis for high-level reporting.
dim_territory	This dimension table provides details about sales

	territories, regions, and countries.
dim_ship_method	This dimension table describes the available shipping methods and their costs.
dim_salesperson	This dimension table contains information about individual sales employees.
dim_product	This dimension table holds descriptive details about each product sold.
dim_date	This dimension table provides a complete calendar with temporal attributes.
dim_costumer	This dimension table contains detailed information about customers.

Relationship between tables is achieved using common table key columns. The following table presents each relationship and their common keys:

Table 1	Table 2	Cardinality
fact_sales_order (customer_id)	dim_costumer (customer_id)	Many to one
fact_sales_order (order_date)	dim_date (date_day)	Many to one
fact_sales_order (product_id)	dim_product (product_id)	Many to one
fact_sales_order (sales_person_id)	dim_salesperson (sales_person_id)	Many to one
fact_sales_order (ship_method_id)	dim_ship_method (ship_method_id)	Many to one
fact_sales_order(territory_id)	dim_territory (territory_id)	Many to one

3. Dashboard

3.1. Sidebar Menu

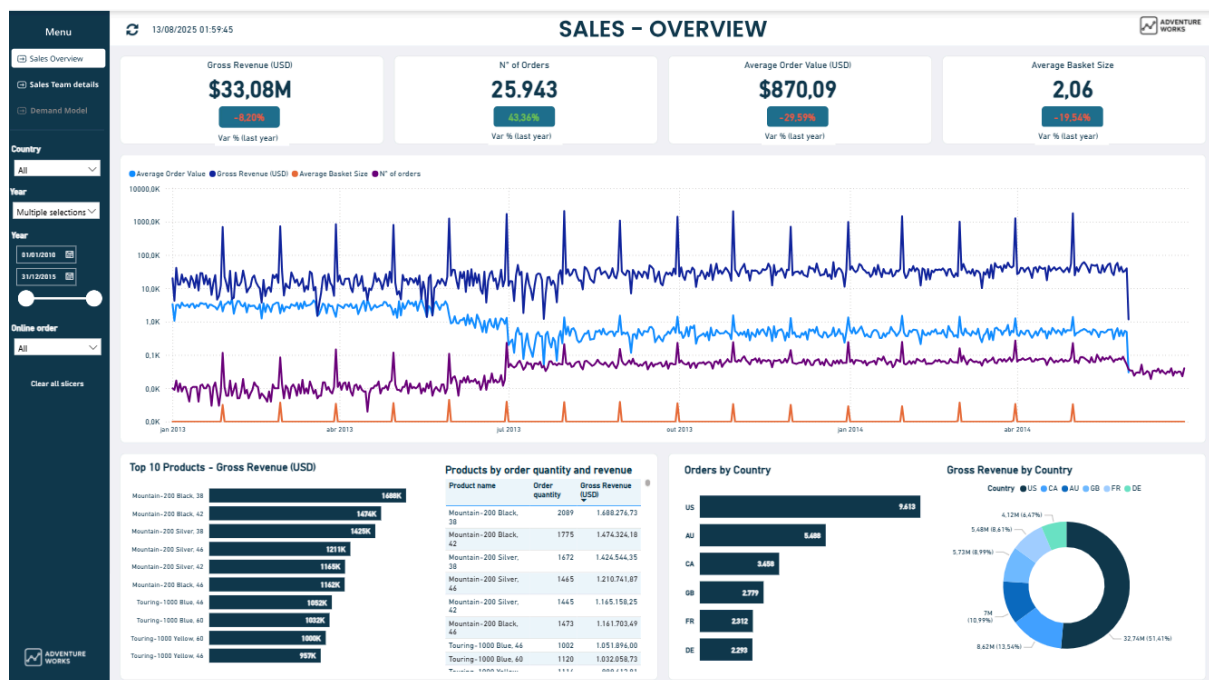
The dashboard's main menu is located at the left side of the page, where we may find the navigation bar (2) and filters (3). Not all options are available in all panels. The dashboard update date is located in the up left corner (5).

2	Navigation Menu Displays navigation buttons for each page of the dashboard
3	Filters Each dashboard page has a slightly different filter options
4	Erase Button Allows users to efficiently reset all applied filters, returning the panel to its original state
5	Refresh Date States the last time the dashboard was updated

Please notice that the data displayed in the visualizations depends on the settings selected in the side menu, such as timeline metric selection and filters.

3.2. Sales Overview

This panel offers a comprehensive overview of the Adventure works key sales metrics and other performance indicators, focusing on gross revenue and orders distribution by location and product.



This page has a total of 9 visualizations, including 4 KPIs, 2 bar charts, 1 line chart, 1 donut chart and 1 table.

Visualization	Type	Description
Gross Revenue (USD)	KPI	This KPI shows the total sales amount in USD for the selected period.
Nº of orders	KPI	This KPI displays the total count of sales orders placed
Average order value (USD)	KPI	This KPI calculates the average revenue per order in USD.
Average Basket Size	KPI	This KPI measures the average number of items per order.
KPI's Line chart	Line chart	This line chart visualizes the trend of all KPIs over time.
Top-10 products - Gross Revenue (USD)	Bar chart	This bar chart ranks the top 10 products by their total revenue.
Product by order quantity and revenue	table	This table provides detailed data for products, including both order quantity and revenue.
Orders by country	Bar chart	This bar chart compares the total number of orders across different countries.
Gross Revenue by country	Donut chart	This donut chart shows the distribution of total revenue by country.

Please notice that the data displayed in the visualizations depends on the settings selected in the side menu, such as timeline metric selection and filters.

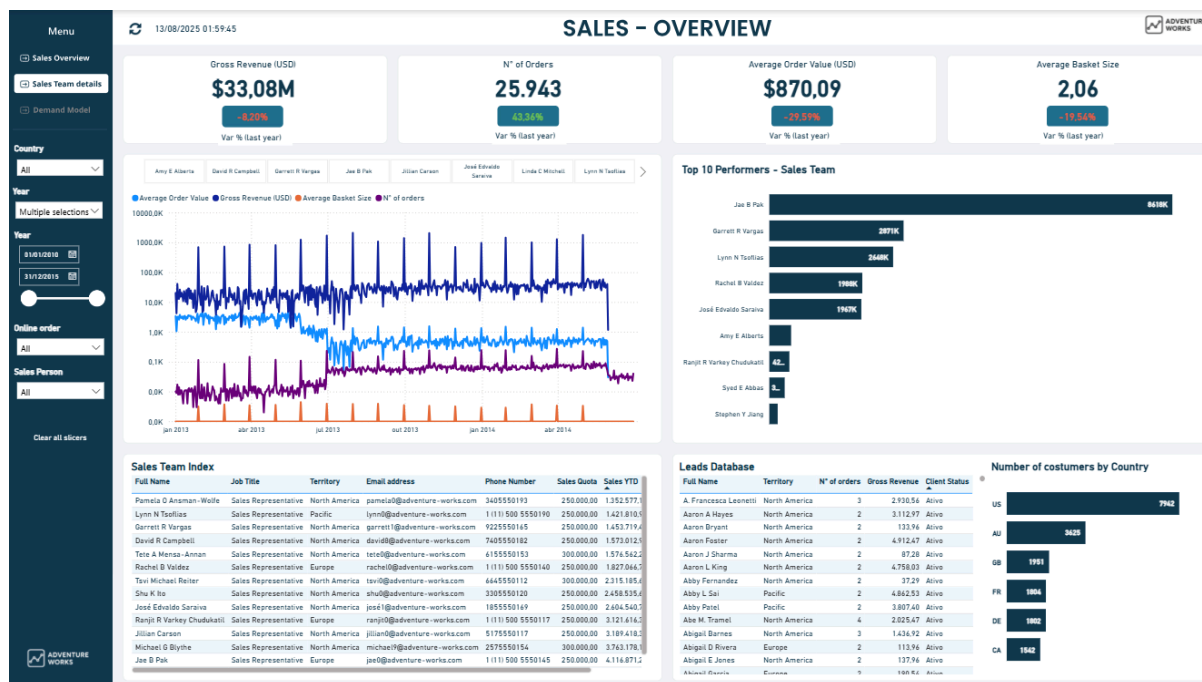
Page Filters

In this page we have the following filters available:

- Country
- Year
- Date (between)
- Online Order

3.3. Sales Team details

This panel offers a comprehensive overview of key performance indicators on Adventure Works sales teams, focusing on sales person performance and client categorization.



This page has a total of 9 visualizations, including 4 KPIs, 2 tables, 2 bar charts and 1 line charts.

Visualization	Type	Description
Gross Revenue (USD)	KPI	This KPI shows the total sales amount in USD for the selected period.
N° of orders	KPI	This KPI displays the total count of sales orders placed
Average order value (USD)	KPI	This KPI calculates the average revenue per order in USD.
Average Basket Size	KPI	This KPI measures the average number of items per order.
KPI's Line chart	Line chart	This line chart visualizes the trend of all KPIs over time.
Top 10 Performers – Sales team	Bar chart	Ranks the top 10 salespeople based on their gross revenue

Sales Team Index	Table	Presents a detailed table of the entire sales team, including key metrics for each salesperson.
Leads Database	Table	Displays a complete table of all potential and pass customers, or leads, with their contact and status information
Number of costumer by country	Bar chart	Compares the total count of customers across different countries to show geographic distribution.

Please notice that the data displayed in the visualizations depends on the settings selected in the side menu, such as timeline metric selection and filters.

Page Filters

In this page, we have six filters available:

- Country
- Year
- Date (between)
- Sales Person
- Online Order

4. Documentation's Changes and Updates

Data	Author	Description
13/08/2025	Alexandre R.S.F	v.1 of the document